# Free-body diagrams



**antocciphysics | TEACHER KEY | CPA**

**Learning target:** Draw only the external forces on one chosen object.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 1, Day 12; 3 course days later.*
Mass 2 kg; new acceleration 5 m/s². At fixed mass, acceleration scales with net force.

**02. Explain a worked example (worked)**
Tension up and weight down; lamp-on-cable force acts on a different object.

**03. A book at rest (model)**
Equal normal up and weight down; table and Earth.

**04. After the hand lets go (practice)**
Weight down, normal up, kinetic friction left; no hand force after release.

**05. A falling ball (reason)**
Weight down longer than upward drag; net force down.

**06. Choose a method (mixed)** *Review: Unit 1, Day 3; 12 course days later.*
v=(6-18)/4=-3 m/s, left. Straight descending graph between the stated endpoints.

**07. Exit check (exit)**
Only Earth’s weight down. Hand no longer contacts; upward velocity not a force.

**08. STRETCH: Apply it another way (stretch)**
Both diagrams can have equal tension and weight. Force balance alone does not tell velocity.

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**Learning target:** Draw only the external forces on one chosen object.

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**01. Bring back an earlier idea (retrieve)** *Review: Unit 1, Day 12; 3 course days later.*
Mass=1/slope≈0.6667 kg; a=3.75 m/s². Resistance subtracts from driving force, producing an offset.

**02. Explain a worked example (worked)**
Tension up and weight down; lamp-on-cable force acts on a different object.

**03. Motionless lamp (model)**
Tension 19.6 Nup; weight 19.6 N down.

**04. A rough ramp (practice)**
Weight vertically down; normal perpendicular out; static friction up slope opposes gravity’s downslope tendency.

**05. Critique the extra arrows (reason)**
Remove velocity force. Show net force separately as a sum, not an additional interaction. Check friction direction and actual sources.

**06. Choose a method (mixed)** *Review: Unit 1, Day 3; 12 course days later.*
v=6 m/s; position increases 6 m each second. x(8)=32 m.

**07. Exit check (exit)**
Weight down, normal up, friction right. No vertical acceleration (equal vertical forces); horizontal acceleration right if friction only horizontal force.

**08. STRETCH: Apply it another way (stretch)**
2 T cos 30°=40 N, so T≈23.1 N. Horizontal components cancel; do not set each tension to 20 N.